

Immutable API-derived series
Hourly calendar; invalid pollutant cells become missing



Chronological 90/10 calendar split
Four training-only validation weeks; bounded configurations



At each weekly origin: isolate past data
Training-only transformations; actual future gases (Perfect Prognosis)



Weekly expanding refit
Refit parameters and scaler
Use all observed pre-origin history



Frozen parameters and scaler
Refresh SARIMAX state / NP context
Apply previously available EWMA bias



168-hour forecast; next origin after 168 hours
Future targets withheld until after prediction



Reveal outcomes; score common observed hours
23 origins; 3,624 hours; 19 complete + 4 partial weeks
Update bias from base residuals for the next origin



Pooled and weekly errors; paired block intervals; seeds
Lead times, controls, components, and timing boundaries